A superquadratic separation between sensitivity and block sensitivity

Rewritten for readability from the OpenAI manuscript of the same title

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Abstract

We disprove the quadratic strengthening of the Sensitivity Conjecture by constructing nonconstant total Boolean functions whose block sensitivity exceeds any prescribed constant multiple of sensitivity squared (Theorem 1.1). Composing one such function with itself then gives, for some fixed $\alpha > 2$, a sequence of functions F_ν with unbounded block sensitivity and $\mathrm{bs}(F_\nu) \geq \mathrm{bs}(F_\nu, 0) \geq s(F_\nu)^\alpha$ (Corollary 1.2).

1 Results, context, and the proof at a glance

1.1 Definitions and statements

For a positive integer n, write $[n] = \{1, \ldots, n\}$. A *total Boolean function* is a map $f : \{0,1\}^n \to \{0,1\}$ defined on every input. If $x \in \{0,1\}^n$ and $B \subseteq [n]$, let x^B be the string obtained from x by flipping exactly the coordinates in B; abbreviate $x^{\{e\}}$ to x^e. We write 0 for the all-zero string, so 0^B is the indicator string of B.

The *sensitivity* of f is

$$s(f,x) = \big|\{e \in [n] : f(x^e) \neq f(x)\}\big|, \qquad s(f) = \max_x s(f,x).$$

A *sensitive block for f at x* is a nonempty set $B \subseteq [n]$ with $f(x^B) \neq f(x)$. The *block sensitivity* $\mathrm{bs}(f,x)$ is the largest number of pairwise disjoint sensitive blocks for f at x, and $\mathrm{bs}(f) = \max_x \mathrm{bs}(f,x)$. In particular $\mathrm{bs}(f) \geq \mathrm{bs}(f,0)$.

Theorem 1.1. *For every real $C > 0$, there are a positive integer n and a nonconstant total Boolean function $f : \{0,1\}^n \to \{0,1\}$ such that*

$$\mathrm{bs}(f) > C\, s(f)^2.$$

More precisely, for every integer $d \geq 1$ the construction below gives a nonconstant total Boolean function f with

$$\frac{\mathrm{bs}(f)}{s(f)^2} \geq \frac{2^d}{4(d+2)^2}.$$

Reading the second assertion. The statement is the source's. "The construction below" depends on d, on a size parameter M, and on an edge labelling; these choices are made in the proof (Section 5.2), not in the statement. The ratio bound is proved for *every* integer $M \geq 9^{d+1}$ and *every* labelling as in Lemma 2.1; it is not claimed for smaller M.

Corollary 1.2. *There exist a real number $\alpha > 2$ and nonconstant total Boolean functions F_ν, for integers $\nu \geq 1$, such that*

$$\mathrm{bs}(F_\nu, 0) \geq s(F_\nu)^\alpha \qquad \textit{for every } \nu, \qquad \mathrm{bs}(F_\nu, 0) \longrightarrow \infty.$$

What the proofs deliver concretely. For every $d \geq 1$, every integer $M \geq 9^{d+1}$ and every labelling as in Lemma 2.1, the proofs produce a function f determined by these three data. The labelling is only shown to exist (by a probabilistic argument), so f is not explicit in the usual sense. It satisfies $f(0) = 0$ and

$$s(f) \leq 2(d+2)M^{d+1}, \qquad \mathrm{bs}(f,0) \geq M^2(2M^2+1)^d;$$

see (10) and (11). The resulting lower bound $2^d/(4(d+2)^2)$ for $\mathrm{bs}(f,0)/s(f)^2$ is below 1 for $d \leq 8$ and equals $512/484 > 1$ at $d = 9$. The Corollary needs a seed with $\mathrm{bs}(f,0) > s(f)^2$ and takes $d = 9$, the smallest d for which this bound guarantees it: with A, B the two displayed bounds at $d = 9$, it uses $B/A^2 = (k/M^2)^9/484 \geq 512/484$, where $k = 2M^2+1$. The exponent α of the Corollary is $\log B/\log A$; it depends on the chosen $M \geq 9^{10}$, is fixed once M is fixed, is not canonical, and is only slightly larger than 2.

1.2 Context (not used in the proofs)

Every sentence of this subsection is background and attribution reproduced from the source. None of it is used in a proof (the proofs are self-contained), and none of it was checked against the cited papers.

The Sensitivity Conjecture asked whether block sensitivity is bounded by a polynomial in sensitivity, uniformly over the number of inputs and the function. The question goes back to Nisan's work on parallel computation, which introduced block sensitivity and already recorded Rubinstein's quadratic separation [Nis89, Section 2]. Nisan and Szegedy explicitly suggested the quadratic inequality $\mathrm{bs}(f) \leq s(f)^2$ [NS94, Section 4]. Rubinstein's construction [Rub95] showed that a polynomial upper bound would need degree at least two. Virza improved the separation to $\mathrm{bs}(f) = \frac{1}{2}s(f)^2 + \frac{1}{2}s(f)$ [Vir11], and Ambainis and Sun obtained $\mathrm{bs}(f) = \frac{2}{3}s(f)^2 - \frac{1}{3}s(f)$ [AS11, Theorem 1]. Huang resolved the polynomial conjecture in 2019: his degree bound, together with Tal's refinement of the degree–block-sensitivity comparison [Tal13], gives $\mathrm{bs}(f) \leq s(f)^4$ for every total Boolean function [Hua19, Theorem 1.5]. Wellens improved the leading constant to $\mathrm{bs}(f) \leq \sqrt{2/3}\, s(f)^4 + 1$ [Wel22, Corollary 4.2]. Theorem 1.1 rules out the quadratic strengthening; it does not conflict with the quartic upper bound, since it only excludes exponent 2.

Relation to spectral sensitivity. According to the source, Meiburg's spectral-sensitivity construction uses tournament-indexed rows of positive conditions and one selected blocking coordinate in each outgoing row [Mei26, Sections 3.1 and 5.2–5.5].[1] The present construction adapts that architecture (rows, tournament, one labelled child per outgoing edge) to *nested* predicates. What is different, again according to the source, is the forbidden-label condition (here Lemma 2.1, proved by a direct union bound) and the use of simultaneous changes of two predicates to control ordinary sensitivity. Nothing from [Mei26] is used as a logical input. Spectral sensitivity $\lambda(f)$ is the operator norm of the adjacency matrix of the graph joining adjacent inputs on which f differs [ABDK+21]. The maximum degree of this graph is $s(f)$, and the operator norm of a symmetric 0/1 matrix is at most its largest row sum, so $\lambda(f) \leq s(f)$. Hence a lower bound on $\mathrm{bs}(f)/\lambda(f)^2$ alone would not give Theorem 1.1.

Other borrowed principles. Taking an OR of disjoint copies to balance the sensitivities at zero-inputs and one-inputs follows Ambainis and Sun [AS11, Section 4, Lemma 1]. Ambainis and Prūsis noted that a seed with block sensitivity greater than sensitivity squared would yield a

[1] In the attribution accompanying this construction, Meiburg credits GPT-5.6-Sol with the gated-tournament ideas and describes his own contributions as exposition, size and parameter optimization, and checking the faithfulness of the Lean formalization.

superquadratic power separation by iteration [AP14, Section 1]; composition as an amplification method is treated systematically by Tal [Tal13]. The two composition facts we need are proved in Lemma 6.1.

1.3 Proof at a glance

The object. Fix a depth $d \geq 1$ and a size parameter M, and put $L = d + 2$, $k = 2M^2 + 1$, $r = \lceil \sqrt{M} \rceil$. For each level $\ell = 0, \ldots, d$ we define a short list of Boolean functions ("predicates") on the same N_ℓ bits, $P_{\ell,1} \geq P_{\ell,2} \geq \cdots \geq P_{\ell,H_\ell}$ pointwise; this chain is what "nested" means. At level 0, $P_{0,q}$ accepts the inputs within Hamming distance $d + 1 - q$ of one of M far-apart centres. A level-ℓ input is cut into kr level-$(\ell - 1)$ inputs ("children"), arranged in k rows of r. The rows are the vertices of a tournament in which every row has exactly M^2 out-neighbours, and each edge $i \to j$ carries a label $a(i, j)$ naming one child of row j. The predicate $P_{\ell,q}$ is an OR of k clauses, one per row. Clause i requires:

- *targets*: the strongest (highest-index) child predicate accepts all r children of row i;
- *gates*: for each of the M^2 edges $i \to j$, a weaker child predicate (which one depends on q) rejects the labelled child of row j.

The final function f is the OR of M disjoint copies of $P_{d,1}$.

Why block sensitivity is large (easy). At the all-zero input every predicate rejects. Flipping, in every child of one row, a block that turns that child on satisfies that row's clause: its targets hold, and its gate children lie in other rows, are untouched, and still reject. Every one of the k rows can be used, so the number of disjoint sensitive blocks at zero is multiplied by $k = 2M^2 + 1$ per level (Lemma 3.2), giving $\mathrm{bs}(f, 0) \geq M^2 k^d$.

Why sensitivity is small (the work). Let $S_{\ell,0}$ (resp. $S_{\ell,1}$) be the largest number of single-bit flips that take some $P_{\ell,q}$ from 0 to 1 (resp. from 1 to 0) at some input. The goal is $S_{\ell,b} \leq 2LM^{\ell+b}$: sensitivity grows by a factor about M per level, so s^2 grows by about M^2, while the block count grows by $2M^2 + 1$. The gain is a factor 2 per level.

- *From* 1 *to* 0. Fix one satisfied clause; a flip must break one of its M^2 gates or r targets. Hence $S_{\ell,1} \leq M^2 S_{\ell-1,0} + r S_{\ell-1,1}$.
- *From* 0 *to* 1. A flip touches one child, and a clause tests each child at most once (Lemma 2.3), so the flip must complete a clause that was exactly one condition short. Call $\mathcal{T}$ the clauses with exactly one failed target, and $\mathcal{G}$ those with exactly one failed gate. The labels force $|\mathcal{T}| \leq 15$ (Lemma 2.1), and each such clause costs $S_{\ell-1,0}$, a lower-order quantity. The tournament forces $|\mathcal{G}| \leq 3$. Repairing a failed gate means turning a child predicate from 1 to 0, which may cost $S_{\ell-1,1}$, the main term. *Heuristic, not used in the proofs:* if all three clauses were charged $S_{\ell-1,1}$, the recurrences would read roughly $S_{\ell,0} \lesssim 3S_{\ell-1,1}$ and $S_{\ell,1} \lesssim M^2 S_{\ell-1,0}$, so the sensitivity bound would gain a factor $3M^2$ every two levels; its square would grow by about $3M^2$ per level, more than the block growth $2M^2 + 1$. The key observation is that for all clauses in $\mathcal{G}$ except at most one (the *exceptional* clause, defined in Step C3 of the proof of Proposition 4.1), the failed gate sits on a child that is a satisfied target of another clause of $\mathcal{G}$. There the strongest child predicate is 1, and by nesting the repair must turn *both* the gate predicate and the strongest predicate to 0. Such flips are counted by a *joint sensitivity* $J_{\ell-1,1}$, which will be tiny. Hence $S_{\ell,0} \leq 16\, S_{\ell-1,0} + S_{\ell-1,1} + 3J_{\ell-1,1}$.

- *Controlling J.* $J_{\ell,b}$ counts flips that change two distinct predicates $P_{\ell,q}, P_{\ell,q'}$ together from b to $1-b$. At level 0 it is zero, since one flip changes a Hamming distance by at most 1 and cannot cross two integer radii (Lemma 3.1). It obeys its own recurrences, $J_{\ell,1} \le M^2 J_{\ell-1,0} + rS_{\ell-1,1}$ and $J_{\ell,0} \le 16\, S_{\ell-1,0} + 3J_{\ell-1,1}$. In the last bound even the exceptional clause is charged $J_{\ell-1,1}$: making a clause hold for two parent indices $q < q'$ forces two child gate predicates to change.

- *Solving the recurrences.* After dividing by $M^{\ell+b}$, the normalised S-quantity u_ℓ and J-quantity v_ℓ satisfy $u_\ell \le (1+\epsilon)u_{\ell-1} + 3v_{\ell-1}$, $v_\ell \le \epsilon u_{\ell-1} + 3v_{\ell-1}$ with $\epsilon \le 2/\sqrt{M}$. The quantity v may triple per level, but it starts at 0 and is fed only by ϵu. The hypothesis $M \ge 9^{d+1}$ gives $\epsilon\, 3^{d+1} \le 2$, which keeps $u_\ell \le 2L$ for all $\ell \le d$ (Lemma 5.1).

- *Balancing.* $P_{d,1}$ has $S_{d,0} \le 2LM^d$ but $S_{d,1} \le 2LM^{d+1}$. The OR of M copies has sensitivity at most $\max\{MS_{d,0}, S_{d,1}\} \le 2LM^{d+1}$ and M times as many blocks.

Then $\mathrm{bs}(f,0)/s(f)^2 \ge M^2k^d/(4L^2M^{2d+2}) = (k/M^2)^d/(4L^2) \ge 2^d/(4L^2)$.

The Corollary. For disjoint composition, sensitivity is submultiplicative and block sensitivity at zero is supermultiplicative (Lemma 6.1). Iterating the seed $d = 9$, for which $\mathrm{bs}(f,0) \ge B > A^2 \ge s(f)^2$, gives $\mathrm{bs}(F_\nu, 0) \ge B^\nu$ and $s(F_\nu) \le A^\nu$.

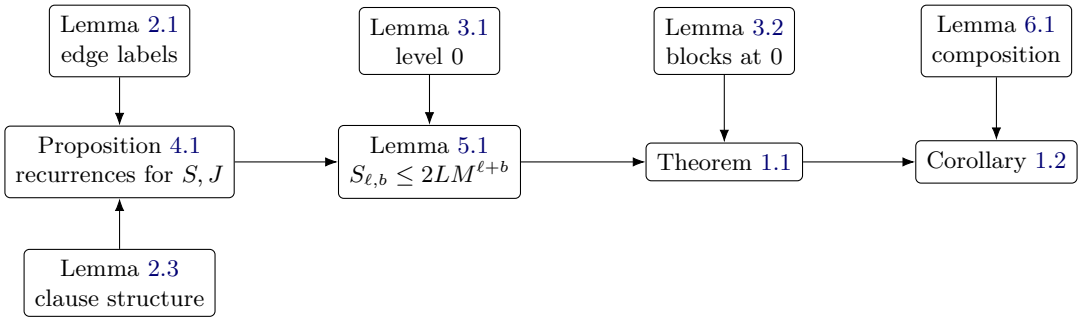


Figure 1: Dependencies. An arrow means "is used in the proof of".

1.4 Notation, and symbols renamed from the source

Each symbol below has one meaning throughout.

$d,\ M,\ L = d+2,\ k = 2M^2+1,\ r = \lceil\sqrt{M}\rceil,\ t = 16$	global parameters
ℓ; $H_\ell = d+1-\ell$; $N_\ell = LM(kr)^\ell$	level; number of predicates; number of bits
$q < q'$	indices of predicates $P_{\ell,q}, P_{\ell,q'}$ at the level ℓ under study (called the "parent" level when its children, at level $\ell-1$, are in play)
$p < p'$	generic predicate indices: at any level in (N0)/(N1), at the child level $\ell-1$ in rule (R)
$H = H_\ell$, $Q = H+1 = H_{\ell-1}$	shorthand at a fixed level $\ell \ge 1$
$i, j \in [k]$; $c \in [r]$; $a(i,j) \in [r]$	rows; positions; edge labels
e	a single coordinate (bit) of an input
$S_{\ell,b}$, $J_{\ell,b}$; S'_b, J'_b	profiles; the same at level $\ell-1$

To keep one meaning per symbol, the following names differ from the source; the mathematics is unchanged. Single coordinates are called e (source: a, which also names the edge labels). The gate indices are $\gamma = Q - q$, $\gamma' = Q - q'$ (source: g, g'). The size of $\mathcal{G}$ is written $|\mathcal{G}|$ (source: m). In Lemma 6.1 the dimensions are n_1, n_2 (source: a, b), the outer string is w (source: y), the blocks are $X_s, Y_{s'}$ (source: C_i, D_j) and their numbers p, p' (source: p, q). The index in the Corollary is ν (source: m).

2 The construction

2.1 Parameters and the tournament

Fix integers $d \geq 1$ and $M \geq 3$, and put

$$L = d + 2, \qquad k = 2M^2 + 1, \qquad r = \lceil \sqrt{M} \rceil, \qquad t = 16.$$

Sections 2–4 only need $M \geq 3$; from Section 5 on we assume $M \geq 9^{d+1}$.

A *tournament* on a vertex set is an orientation of every pair of distinct vertices: for $i \neq j$ exactly one of $i \to j$ and $j \to i$ holds. We fix a tournament on $[k]$ in which every vertex has outdegree exactly M^2. One exists: identify $[k]$ with $\mathbb{Z}/k$ and let $i \to j$ when $j - i \bmod k \in \{1, \dots, M^2\}$. For $i \neq j$ the residues $j - i$ and $i - j$ are nonzero and sum to $k = 2M^2 + 1$, so exactly one of them lies in $\{1, \dots, M^2\}$.

2.2 The edge labelling

The next lemma is used exactly once, in Step C2 of the proof of Proposition 4.1, to show $|\mathcal{T}| < t$.

Lemma 2.1. *The edges can be labelled by numbers $a(i,j) \in [r]$ so that there is no set $I \subseteq [k]$ of size t admitting choices $m_j \in [r]$, $j \in I$, with*

$$a(i,j) = m_j \qquad \text{for every edge } i \to j \text{ within } I.$$

Proof. Choose all edge labels independently and uniformly from $[r]$.

Step 1 (one configuration). Fix I with $|I| = t = 16$ and fix $(m_j)_{j \in I}$. Every unordered pair of vertices of I carries exactly one edge, so there are $\binom{16}{2} = 120$ edges within I. The event "$a(i,j) = m_j$ for all of them" is the intersection of 120 independent events of probability $1/r$, so it has probability r^{-120}.

Step 2 (union bound). There are $\binom{k}{t} \leq k^{16}$ sets I and r^{16} choices of (m_j). The probability that some configuration occurs is therefore at most

$$k^{16}\, r^{16}\, r^{-120} = k^{16} r^{-104} \leq (3M^2)^{16} (\sqrt{M})^{-104} = 3^{16} M^{32} M^{-52} = 3^{16} M^{-20} \leq 3^{16}\, 3^{-20} < 1.$$

Here we used $k = 2M^2 + 1 \leq 3M^2$, $r \geq \sqrt{M}$, and $M \geq 3$.

Step 3. With positive probability no configuration occurs, so a labelling with the required property exists. □

Fix such a labelling for the remainder of the paper.

2.3 Levels and predicates

For $0 \leq \ell \leq d$, set

$$H_\ell = d + 1 - \ell, \qquad N_\ell = LM(kr)^\ell.$$

We define predicates $P_{\ell,q} : \{0,1\}^{N_\ell} \to \{0,1\}$ for $1 \leq q \leq H_\ell$. Level 0 has $d + 1$ predicates and level d has exactly one, $P_{d,1}$. We say that $P_{\ell,q}$ *accepts* y if $P_{\ell,q}(y) = 1$ and *rejects* y otherwise.

Level 0. Partition $[LM]$ into M blocks $E_1, \dots, E_M$, each of size L. Let $z_j = 0^{E_j}$ be the string that is one exactly on E_j (the j-th *centre*). Write $\mathrm{dist}(x, z)$ for Hamming distance and set

$$D(x) = \min_{j \in [M]} \mathrm{dist}(x, z_j), \qquad P_{0,q}(x) = \mathbf{1}\{D(x) \le H_0 - q\} = \mathbf{1}\{D(x) \le d + 1 - q\}.$$

So $P_{0,q}$ has radius $d + 1 - q$, running from d (for $q = 1$) down to 0 (for $q = d + 1$). Two distinct centres differ exactly on $E_j \cup E_{j'}$, so their distance is $2L = 2d + 4$.

Level $\ell \ge 1$. Suppose $1 \le \ell \le d$ and that level $\ell - 1$ is defined. Since $N_\ell = kr\, N_{\ell-1}$, an input $x \in \{0,1\}^{N_\ell}$ can be partitioned into kr disjoint *children* $x_{j,c} \in \{0,1\}^{N_{\ell-1}}$, arranged in k rows $j \in [k]$, each with r positions $c \in [r]$. "Child (j, c)" refers both to the string $x_{j,c}$ and to its set of $N_{\ell-1}$ coordinates. Put

$$H = H_\ell, \qquad Q = H + 1 = H_{\ell-1}.$$

For $i \in [k]$ and $q \in [H]$, *clause i at index q* is the conjunction of the following $r + M^2$ conditions:

$$\begin{aligned} P_{\ell-1,Q}(x_{i,c}) &= 1 \quad \text{for all } c \in [r] \qquad (\textit{target conditions}), \\ P_{\ell-1,Q-q}(x_{j,a(i,j)}) &= 0 \quad \text{for all } j \text{ with } i \to j \quad (\textit{gate conditions}). \end{aligned} \tag{1}$$

Define $P_{\ell,q}(x) = 1$ if clause i at index q is satisfied for at least one $i \in [k]$, and $P_{\ell,q}(x) = 0$ otherwise. The indices are valid: for $q \in [H]$ we have $1 \le Q - q \le H < Q = H_{\ell-1}$. The targets use the highest child index Q and do not depend on q; the gates use the child index $Q - q$, which is always strictly smaller than Q. Figure 2 depicts one clause.

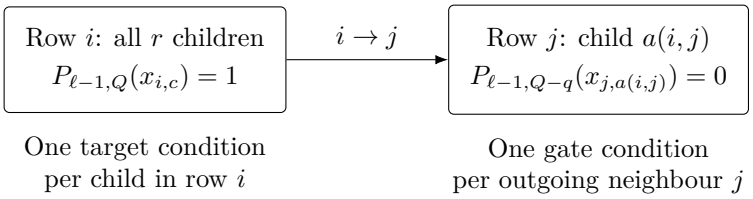


Figure 2: One recursive clause. The edge label selects a single child in each outgoing neighbour's row. Raising q strengthens the gates; all targets use the strongest child predicate Q.

Remark 2.2 (Smallest example). For $d = 1$: level 0 has $P_{0,1}$ (radius 1) and $P_{0,2}$ (radius 0). At level 1 we have $H = 1$, $Q = 2$, and only $q = 1$. Clause i says that every child of row i *equals* a centre, and that for each edge $i \to j$ the labelled child of row j is at distance at least 2 from every centre.

2.4 Structure of the clauses

The following lemma collects the elementary facts used in every counting argument of Section 4. For a predicate list at level ℓ and indices $p < p'$, nesting $P_{\ell,p} \ge P_{\ell,p'}$ is used in two equivalent forms, which we name:
(N1) $P_{\ell,p'}(y) = 1 \implies P_{\ell,p}(y) = 1$ (acceptance passes to every *smaller* index);
(N0) $P_{\ell,p}(y) = 0 \implies P_{\ell,p'}(y) = 0$ (rejection passes to every *larger* index).
Nesting concerns the predicate index, not the input bits: the predicates are not monotone in x.

Lemma 2.3. (a) (Nesting.) *For every $0 \le \ell \le d$, every $y \in \{0,1\}^{N_\ell}$ and all $1 \le p < p' \le H_\ell$, $P_{\ell,p}(y) \ge P_{\ell,p'}(y)$; that is,* (N1) *and* (N0) *hold. Moreover, for $\ell \ge 1$, $i \in [k]$ and $q < q'$ in $[H_\ell]$: every condition of clause i that holds at index q' also holds at index q (the gate for a given edge at index q' implies the gate for the same edge at index q; the targets are identical). In particular, clause i satisfied at index q' implies clause i satisfied at index q.*

(b) (Distinct children.) *Let $\ell \geq 1$ and $i \in [k]$. The children tested by clause i are the r target children (i, c), $c \in [r]$, and the M^2 gate children $(j, a(i, j))$, $i \to j$. These $r + M^2$ children are pairwise distinct and do not depend on the index q.*

(c) (One flip, one condition.) *Let $\ell \geq 1$, $x \in \{0,1\}^{N_\ell}$, and let e be a coordinate, lying in child (j, c). For every clause i and index q, every condition of clause i that does not test child (j, c) has the same truth value at x and at x^e; and at most one condition of clause i at index q tests child (j, c).*

(d) (Repair.) *In the situation of (c), suppose clause i at index q is false at x and true at x^e. Then at x clause i has* exactly one *failed condition at index q, and this condition tests the child containing e.*

Proof. (a) By induction on ℓ. At level 0, $p < p'$ gives $d + 1 - p > d + 1 - p'$, so $D(y) \leq d + 1 - p'$ implies $D(y) \leq d + 1 - p$. Let $\ell \geq 1$ and assume nesting at level $\ell - 1$. Let $q < q'$. The targets of clause i do not involve the index. For a gate, $Q - q' < Q - q$, so if the gate holds at index q', i.e. $P_{\ell-1,Q-q'} = 0$ on its child, then (N0) at level $\ell - 1$ gives $P_{\ell-1,Q-q} = 0$ on that child, i.e. the gate holds at index q. Hence clause i at q' implies clause i at q, and taking the OR over i gives $P_{\ell,q'} \leq P_{\ell,q}$.

(b) Target children lie in row i at distinct positions. Gate children lie in rows $j \neq i$, and there is exactly one gate per out-neighbour j, so different gates lie in different rows. The list does not mention q.

(c) The strings x and x^e agree on every child other than (j, c), and each condition depends only on the child it tests. The second assertion is (b).

(d) Clause i is false at x, so it has at least one failed condition there. Every condition failed at x holds at x^e, hence changes truth value, hence tests the child containing e by (c); and at most one condition does. □

3 Profiles, the base level, and blocks at zero

3.1 Sensitivity profiles

For $0 \leq \ell \leq d$ and $b \in \{0, 1\}$, define

$$S_{\ell,b} = \max_{\substack{1 \leq q \leq H_\ell,\ x \in \{0,1\}^{N_\ell} \\ P_{\ell,q}(x) = b}} \big|\{e \in [N_\ell] : P_{\ell,q}(x^e) = 1 - b\}\big|,$$

$$J_{\ell,b} = \max_{\substack{1 \leq q < q' \leq H_\ell,\ x \in \{0,1\}^{N_\ell} \\ P_{\ell,q}(x) = P_{\ell,q'}(x) = b}} \big|\{e \in [N_\ell] : P_{\ell,q}(x^e) = P_{\ell,q'}(x^e) = 1 - b\}\big|.$$

Thus $S_{\ell,0}$ bounds the number of flips taking any one predicate of level ℓ from 0 to 1, and $S_{\ell,1}$ those from 1 to 0. $J_{\ell,b}$ bounds the number of flips that move two *distinct* predicates of level ℓ, evaluated at the same input, simultaneously from b to $1 - b$. A maximum over no cases is defined to be zero; in particular $J_{d,0} = J_{d,1} = 0$, since $H_d = 1$.

3.2 The base level

The next lemma supplies the initial values for the induction in Lemma 5.1.

Lemma 3.1. *The initial sensitivity profiles satisfy*

$$S_{0,0} \le L, \qquad S_{0,1} \le LM, \qquad J_{0,0} = J_{0,1} = 0. \tag{2}$$

Proof. Step 1 ($S_{0,0} \le L$). Fix q, put $R = d + 1 - q \in \{0, \dots, d\}$, and let x satisfy $P_{0,q}(x) = 0$, i.e. $D(x) \ge R+1$. Call a coordinate e *successful* if $P_{0,q}(x^e) = 1$. For a successful e, choose a centre z_j with $\mathrm{dist}(x^e, z_j) \le R$. Then $\mathrm{dist}(x, z_j) \le R+1$ by the triangle inequality, and $\mathrm{dist}(x, z_j) \ge D(x) \ge R+1$. So $\mathrm{dist}(x, z_j) = R + 1$ and flipping e decreased the distance to z_j. Hence e is one of the $R + 1$ coordinates in which x and z_j differ.

The same centre serves every successful e: at most one centre lies within distance $R + 1$ of x, because two distinct centres within distance $R + 1 \le d + 1$ of x would be at mutual distance at most $2d + 2 < 2d + 4 = 2L$, whereas distinct centres are at distance $2L$. So all successful coordinates lie in one set of $R + 1$ coordinates; their number is at most $R + 1 \le d + 1 \le L$. As q and x were arbitrary, $S_{0,0} \le L$.

Step 2 ($S_{0,1} \le LM$). This is the number of input bits.

Step 3 (D is 1-Lipschitz). For adjacent inputs x, y and every j, $\mathrm{dist}(x, z_j) \le \mathrm{dist}(y, z_j) + 1$; taking minima gives $D(x) \le D(y) + 1$, and by symmetry $|D(x) - D(y)| \le 1$.

Step 4 ($J_{0,b} = 0$). Let $q < q'$, with radii $R = d + 1 - q$ and $R' = d + 1 - q'$, so $R \ge R' + 1$. If both predicates are 0 at x and both are 1 at x^e, then $D(x) \ge R + 1 \ge R' + 2$ and $D(x^e) \le R'$, a change of at least 2, contradicting Step 3. If both are 1 at x and both are 0 at x^e, then $D(x) \le R'$ and $D(x^e) \ge R + 1 \ge R' + 2$, again impossible. So no coordinate is counted and $J_{0,0} = J_{0,1} = 0$. □

3.3 Disjoint blocks at zero

The next lemma gives the block-sensitivity lower bound in the proof of Theorem 1.1. It concerns only the all-zero input.

Lemma 3.2. *At every level $0 \le \ell \le d$, all predicates are zero at the all-zero input. There are Mk^ℓ pairwise disjoint nonempty blocks of coordinates, each of size Lr^ℓ, such that flipping any one block at zero makes every predicate at that level equal to one.*

Proof. By induction on ℓ.

Level 0. Every centre has exactly L ones, so $D(0) = L > d$, and every radius is at most d; hence all $P_{0,q}(0) = 0$. The M blocks E_j are disjoint, nonempty, of size L, and $0^{E_j} = z_j$ has $D(z_j) = 0$, so every $P_{0,q}(0^{E_j}) = 1$.

Level $\ell \ge 1$, assuming level $\ell - 1$.

Step 1 (all predicates reject 0). All children of the all-zero input are all-zero, so by induction $P_{\ell-1,Q}$ rejects each of them. Every clause has $r \ge 1$ targets, all failing. Hence $P_{\ell,q}(0) = 0$ for every q.

Step 2 (the blocks). Let $B_1, \dots, B_{Mk^{\ell-1}} \subseteq [N_{\ell-1}]$ be the block family of level $\ell - 1$. For a row $i \in [k]$ and an index h, let $B_{i,h}$ be the union, over the r children of row i, of the copy of B_h inside that child.

Step 3 (each $B_{i,h}$ works). In $0^{B_{i,h}}$, each child of row i equals 0^{B_h}, on which every level-$(\ell - 1)$ predicate is 1 by induction; in particular $P_{\ell-1,Q}$ accepts, so all targets of clause i hold. All gate children of clause i lie in rows other than i, are still all-zero, and are rejected by every level-$(\ell - 1)$ predicate, so all gates of clause i hold at every index q. Thus clause i is satisfied at every index and $P_{\ell,q}(0^{B_{i,h}}) = 1$ for all q.

Step 4 (disjointness and counts). For $i \ne i'$ the blocks lie in different rows. For fixed i and $h \ne h'$, they are disjoint inside each child because $B_h \cap B_{h'} = \emptyset$. There are $k \cdot Mk^{\ell-1} = Mk^\ell$ blocks, each nonempty of size $r \cdot Lr^{\ell-1} = Lr^\ell$. □

4 The sensitivity recurrences

This is the core of the paper. The estimates hold at *every* input, not only near the all-zero input. They feed Lemma 5.1.

Proposition 4.1. *For $1 \le \ell \le d$, write $S'_b = S_{\ell-1,b}$ and $J'_b = J_{\ell-1,b}$. Then*

$$\begin{aligned} S_{\ell,1} &\le M^2 S'_0 + r S'_1, & J_{\ell,1} &\le M^2 J'_0 + r S'_1, \\ S_{\ell,0} &\le t S'_0 + S'_1 + 3J'_1, & J_{\ell,0} &\le t S'_0 + 3J'_1. \end{aligned} \tag{3}$$

Plan. Steps A and B treat flips away from the value 1: fix one satisfied clause and count the flips that break it. Step C analyses an input where a predicate is 0: it identifies the clauses that are one condition short (the lists $\mathcal{T}$ and $\mathcal{G}$) and bounds their number. Steps D and E then charge each successful flip to a clause in these lists.

Proof. Fix ℓ, and put $H = H_\ell$, $Q = H + 1 = H_{\ell-1}$ as in (1). Given an input x and a coordinate e, "before" means at x and "after" means at x^e. "The value of $P_{\ell-1,p}$ on child (j,c)" means $P_{\ell-1,p}(x_{j,c})$ before and $P_{\ell-1,p}((x^e)_{j,c})$ after.

Counting rule (R). Fix a child (j,c). The coordinates e of x inside this child correspond bijectively to the coordinates of the string $y = x_{j,c}$, and flipping e replaces y by y with that coordinate flipped while leaving all other children unchanged. Applying the definitions of $S_{\ell-1,b}$ and $J_{\ell-1,b}$ to the input y gives, for $b \in \{0,1\}$:

(R1) for every $p \in [Q]$, at most S'_b coordinates e in child (j,c) change $P_{\ell-1,p}$ on that child from b to $1-b$;

(R2) for all $p < p'$ in $[Q]$, at most J'_b coordinates e in child (j,c) change *both* $P_{\ell-1,p}$ and $P_{\ell-1,p'}$ on that child from b to $1-b$.

A child may be tested by several clauses. This is harmless: we always bound a set of coordinates by a union of sets, one per clause.

Step A: $S_{\ell,1} \le M^2 S'_0 + r S'_1$. Fix $q \in [H]$ and x with $P_{\ell,q}(x) = 1$, and fix one clause i satisfied at x at index q. Let e be a coordinate with $P_{\ell,q}(x^e) = 0$. Then clause i is false after, so some condition of clause i holds before and fails after; by Lemma 2.3(c) it tests the child containing e.

- If it is a gate, then $P_{\ell-1,Q-q}$ on that gate child changes from 0 to 1. By (R1) each gate child contains at most S'_0 such e, and there are M^2 gate children.
- If it is a target, then $P_{\ell-1,Q}$ on that target child changes from 1 to 0. By (R1) each target child contains at most S'_1 such e, and there are r target children.

So at most $M^2 S'_0 + r S'_1$ coordinates e qualify. As q and x were arbitrary, the bound on $S_{\ell,1}$ follows.

Step B: $J_{\ell,1} \le M^2 J'_0 + r S'_1$. Assume $H \ge 2$ (otherwise $J_{\ell,1} = 0$). Fix $q < q'$ in $[H]$ and x with $P_{\ell,q}(x) = P_{\ell,q'}(x) = 1$. Fix one clause i satisfied at x at the *larger* index q'. By Lemma 2.3(a) it is satisfied at x at index q too. Put

$$\gamma = Q - q, \qquad \gamma' = Q - q', \qquad 1 \le \gamma' < \gamma \le Q - 1.$$

So before the flip, on every gate child of clause i,

$$P_{\ell-1,\gamma'} = 0 \quad \text{(gate holds at index } q'\text{)}, \qquad P_{\ell-1,\gamma} = 0 \quad \text{(by (N0), as } \gamma' < \gamma\text{)}. \tag{4}$$

Let e be a coordinate with $P_{\ell,q}(x^e) = P_{\ell,q'}(x^e) = 0$. We use only $P_{\ell,q}(x^e) = 0$. So the count below bounds the set $\{e : P_{\ell,q}(x^e) = 0\}$, which contains the set in the definition of $J_{\ell,1}$; an upper bound

for the larger set suffices. Since $P_{\ell,q}(x^e) = 0$, every clause, in particular clause i, is false after at index q. Since clause i at index q was true before, one of its conditions at index q holds before and fails after, and it tests the child containing e (Lemma 2.3(c)).

- If it is a target: exactly as in Step A, at most rS_1' coordinates in total.
- If it is the gate on some gate child: after the flip the gate at index q fails, i.e. $P_{\ell-1,\gamma} = 1$ on that child, and then (N1) gives $P_{\ell-1,\gamma'} = 1$ on that child. Together with (4), both $P_{\ell-1,\gamma'}$ and $P_{\ell-1,\gamma}$ change from 0 to 1. By (R2) each gate child contains at most J_0' such e, and there are M^2 gate children.

So at most $M^2 J_0' + rS_1'$ coordinates qualify, which is the bound on $J_{\ell,1}$. Steps A and B give the first line of (3).

Step C: the candidate clauses at a zero. Fix $q \in [H]$ and x with $P_{\ell,q}(x) = 0$. In Step C, "fails" and "holds" refer to the input x and the index q. Define

$$\mathcal{T} = \{\, i \in [k] : \text{clause } i \text{ fails exactly one target and no gate} \,\},$$
$$\mathcal{G} = \{\, i \in [k] : \text{clause } i \text{ fails exactly one gate and no target} \,\}.$$

These sets depend only on x and q, not on any flipped coordinate. For $W \in \mathcal{T} \cup \mathcal{G}$ let $\kappa(W)$ be the child tested by the unique failed condition of clause W.

C1 (every successful flip repairs a candidate). Let e be a coordinate and let W be a clause satisfied at x^e at index q. Then $W \in \mathcal{T} \cup \mathcal{G}$ and e lies in child $\kappa(W)$.

Proof. Since $P_{\ell,q}(x) = 0$, clause W is false at x. By Lemma 2.3(d), clause W has exactly one failed condition at x, and it tests the child containing e. That condition is a target (then $W \in \mathcal{T}$) or a gate (then $W \in \mathcal{G}$).

C2 ($|\mathcal{T}| \le t - 1$). For $j \in \mathcal{T}$ let $m_j \in [r]$ be the position of its unique failed target, so $\kappa(j) = (j, m_j)$ and $P_{\ell-1,Q}(x_{j,c}) = 1$ for all $c \neq m_j$. We claim:

$$i, j \in \mathcal{T},\ i \to j \quad \Longrightarrow \quad a(i,j) = m_j.$$

Indeed, clause i fails no gate, so its gate for the edge $i \to j$ holds: $P_{\ell-1,Q-q}(x_{j,a(i,j)}) = 0$. Since $Q - q < Q$, (N0) gives $P_{\ell-1,Q}(x_{j,a(i,j)}) = 0$. That is, the target of clause j at position $a(i,j)$ fails. The only failed target of clause j is at position m_j, so $a(i,j) = m_j$.

If $|\mathcal{T}| \ge t$, pick $I \subseteq \mathcal{T}$ with $|I| = t$. The choices $(m_j)_{j \in I}$ satisfy $a(i,j) = m_j$ for every edge $i \to j$ within I, contradicting Lemma 2.1. Hence $|\mathcal{T}| \le t - 1$.

C3 (structure of $\mathcal{G}$).

(i) *If $i, j \in \mathcal{G}$ and $i \to j$, then the gate of clause i for the edge $i \to j$ fails, and on its child $(j, a(i,j))$ we have $P_{\ell-1,Q} = 1$ and $P_{\ell-1,Q-q} = 1$.* Since $j \in \mathcal{G}$, clause j fails no target, so $P_{\ell-1,Q}(x_{j,c}) = 1$ for every c, in particular for $c = a(i,j)$. Since $Q - q < Q$, (N1) gives $P_{\ell-1,Q-q}(x_{j,a(i,j)}) = 1$, which is the failure of that gate.

(ii) *Each $i \in \mathcal{G}$ has at most one out-neighbour in $\mathcal{G}$.* By (i), distinct out-neighbours in $\mathcal{G}$ give distinct failed gates of clause i, and clause i has exactly one failed gate.

(iii) $|\mathcal{G}| \le 3$. The tournament restricted to $\mathcal{G}$ has $\binom{|\mathcal{G}|}{2}$ edges. Each edge leaves exactly one vertex, so by (ii) the number of edges is at most $|\mathcal{G}|$. Thus $|\mathcal{G}|(|\mathcal{G}| - 1)/2 \le |\mathcal{G}|$, which forces $|\mathcal{G}| \le 3$.

(iv) *At most one $i \in \mathcal{G}$ has no out-neighbour in $\mathcal{G}$.* Of two distinct vertices of $\mathcal{G}$, the edge between them leaves one of them.

Call a clause of $\mathcal{G}$ with no out-neighbour in $\mathcal{G}$ *exceptional*; by (iv) there is at most one. Every non-exceptional $i \in \mathcal{G}$ has a unique out-neighbour $j \in \mathcal{G}$; by (i) its unique failed gate is the one for $i \to j$, so $\kappa(i) = (j, a(i,j))$ and

$$P_{\ell-1,Q-q} = P_{\ell-1,Q} = 1 \quad \text{on child } \kappa(i) \text{ before the flip.} \tag{5}$$

(Examples: if $\mathcal{G} = \{i, j\}$ with $i \to j$, then j is exceptional and i is not; if $|\mathcal{G}| = 3$, the three vertices form a directed cycle and none is exceptional.)

Step D: $S_{\ell,0} \le tS'_0 + S'_1 + 3J'_1$. Keep q, x, $\mathcal{T}$, $\mathcal{G}$, κ from Step C. For $W \in \mathcal{T} \cup \mathcal{G}$ define

$$\Phi(W) = \{\, e \text{ in child } \kappa(W) : \text{clause } W \text{ is satisfied at } x^e \text{ at index } q \,\}.$$

If $P_{\ell,q}(x^e) = 1$, some clause W is satisfied at x^e, and by C1 $e \in \Phi(W)$. Hence

$$\{e : P_{\ell,q}(x^e) = 1\} \subseteq \bigcup_{W \in \mathcal{T} \cup \mathcal{G}} \Phi(W),$$

and it suffices to bound each $|\Phi(W)|$. Let $e \in \Phi(W)$; then the unique failed condition of clause W holds after the flip.

D1. $W = j \in \mathcal{T}$: the target on child (j, m_j) is repaired, so $P_{\ell-1,Q}$ on that child changes from 0 to 1. By (R1), $|\Phi(j)| \le S'_0$.

D2. $W = i \in \mathcal{G}$ non-exceptional. Recall from C3: i has a unique out-neighbour j in $\mathcal{G}$; the only failed condition of clause i is its gate for $i \to j$; the gate child $\kappa(i) = (j, a(i, j))$ is also a target child of clause j, all of whose targets hold because $j \in \mathcal{G}$. This gave (5): both $P_{\ell-1,Q-q}$ and $P_{\ell-1,Q}$ equal 1 on $\kappa(i)$ before. After the flip the gate on $\kappa(i)$ holds, i.e. $P_{\ell-1,Q-q} = 0$ there, and then (N0) gives $P_{\ell-1,Q} = 0$ there. So both predicates change from 1 to 0. The indices $Q - q < Q$ are distinct because $q \ge 1$, and both lie in $[Q]$. By (R2), $|\Phi(i)| \le J'_1$.

D3. $W = i \in \mathcal{G}$ exceptional: the failed gate on $\kappa(i)$ is repaired, so $P_{\ell-1,Q-q}$ on that child changes from 1 to 0. By (R1), $|\Phi(i)| \le S'_1$. (Here nothing is known about $P_{\ell-1,Q}$ on $\kappa(i)$, so (R2) is not available.)

By C2 there are at most $t-1$ clauses of type D1; by C3 there are at most 3 of type D2 and at most 1 of type D3. Therefore the number of successful coordinates is at most $(t-1)S'_0 + 3J'_1 + S'_1 \le tS'_0 + S'_1 + 3J'_1$.

A side effect that needs no accounting. A flip counted in D2 turns $P_{\ell-1,Q}$ to 0 on a child of row j, so it breaks a target of clause j. This is not used and does no harm: $\Phi(i)$ only records that clause i holds after the flip, and one satisfied clause is all that $P_{\ell,q}(x^e) = 1$ requires.

Step E: $J_{\ell,0} \le tS'_0 + 3J'_1$. Assume $H \ge 2$. Fix $q < q'$ in $[H]$ and x with $P_{\ell,q}(x) = P_{\ell,q'}(x) = 0$. Form $\mathcal{T}$, $\mathcal{G}$, κ exactly as in Step C for this x and the *smaller* index q; C1–C3 apply. Put $\gamma = Q - q$, $\gamma' = Q - q'$, so $1 \le \gamma' < \gamma \le Q - 1$. For $W \in \mathcal{T} \cup \mathcal{G}$ define

$$\Phi'(W) = \{\, e \text{ in child } \kappa(W) : \text{clause } W \text{ is satisfied at } x^e \text{ at index } q' \,\}.$$

These finitely many sets are determined by x, q, q' alone.

E0 (reduction). Let e satisfy $P_{\ell,q}(x^e) = P_{\ell,q'}(x^e) = 1$. Choose a clause W satisfied at x^e at index q'; it exists because $P_{\ell,q'}(x^e) = 1$. By Lemma 2.3(a), W is also satisfied at x^e at index q. By C1, $W \in \mathcal{T} \cup \mathcal{G}$ and e lies in $\kappa(W)$. So $e \in \Phi'(W)$. The witness W may vary with e; it serves only to show that e lies in at least one of the fixed sets $\Phi'(W)$. Hence

$$\{e : P_{\ell,q}(x^e) = P_{\ell,q'}(x^e) = 1\} \subseteq \bigcup_{W \in \mathcal{T} \cup \mathcal{G}} \Phi'(W),$$

and it suffices to bound each $|\Phi'(W)|$ and add.

E1 ($W = j \in \mathcal{T}$). By Lemma 2.3(a), $\Phi'(j) \subseteq \Phi(j)$, so $|\Phi'(j)| \le S'_0$ by D1.

E2 ($W = i \in \mathcal{G}$, exceptional or not). The child $\kappa(i)$ is the gate child of the unique gate of clause i that fails at x at index q. Hence, before the flip, $P_{\ell-1,\gamma} = 1$ on $\kappa(i)$, and (N1) gives $P_{\ell-1,\gamma'} = 1$ on $\kappa(i)$ as well. Let $e \in \Phi'(i)$. After the flip, clause i holds at index q'. Its gate for the same edge tests the same child $\kappa(i)$ (Lemma 2.3(b)), so $P_{\ell-1,\gamma'} = 0$ on $\kappa(i)$ after, and (N0) gives $P_{\ell-1,\gamma} = 0$ there. Thus the two distinct predicates $P_{\ell-1,\gamma'}$ and $P_{\ell-1,\gamma}$ both change from 1 to 0 on $\kappa(i)$, and (R2) gives $|\Phi'(i)| \le J'_1$.

Why no S'_1 term appears here. In D2 the second predicate was $P_{\ell-1,Q}$, whose initial value 1 came from the row of $\kappa(i)$ belonging to $\mathcal{G}$; this was unavailable for the exceptional clause, which is why D3 used S'_1. In E2 the second predicate is $P_{\ell-1,\gamma'}$, supplied by the second parent index q'. Its initial value 1 follows from nesting alone, with no information about targets. So E2 covers the exceptional clause too.

E3 (sum). With $|\mathcal{T}| \le t-1$ and $|\mathcal{G}| \le 3$, the number of joint-success coordinates is at most $(t-1)S'_0 + 3J'_1 \le tS'_0 + 3J'_1$.

Step F: conclusion. Steps A, B, D, E hold for every admissible input and every index or pair of indices, so taking maxima gives (3). If $H = 1$ there is no pair $q < q'$, and $J_{\ell,0} = J_{\ell,1} = 0$ by the empty-maximum convention, so the two joint inequalities hold trivially. □

Remark 4.2 (The source's formulation of Step E). The source adds two observations which are true but not needed for the count above. First, a witness W as in E0 had exactly one failed condition at x at index q' as well: since $P_{\ell,q'}(x) = 0$, clause W is false at x at index q' and true at x^e, so Lemma 2.3(d) applies at index q'. That condition tests the child containing e, i.e. $\kappa(W)$. Second, a clause of $\mathcal{T} \cup \mathcal{G}$ with two or more failed conditions at index q' is therefore never a witness; for it $\Phi'(W) = \emptyset$, which only helps the upper bound.

5 Quantitative bounds and the proof of Theorem 1.1

5.1 Solving the recurrences

The dominant terms $M^2 S'_0$ and S'_1 in Proposition 4.1 suggest normalising $S_{\ell,b}$ by $M^{\ell+b}$. The next lemma is used in the proof of Theorem 1.1 at $\ell = d$.

Lemma 5.1. *Suppose that $d \ge 1$ and $M \ge 9^{d+1}$. For every $0 \le \ell \le d$ and $b \in \{0,1\}$, the predicates of the construction satisfy*

$$S_{\ell,b} \le 2LM^{\ell+b}.$$

Plan. Normalise the four recurrences into two (Step 2), dominate the normalised quantities by explicit comparison sequences (Step 4), and bound those by induction (Step 5). The hypothesis $M \ge 9^{d+1}$ is used only in Step 3 (as $\sqrt{M} \ge 9$ and $\sqrt{M} \ge 3^{d+1}$) to obtain $\epsilon \le 2/3^{d+1}$; that bound is used only at the end of Step 5.

Proof. Step 1 (normalised quantities). Define

$$u_\ell = \max_{b\in\{0,1\}} \frac{S_{\ell,b}}{M^{\ell+b}}, \qquad v_\ell = \max_{b\in\{0,1\}} \frac{J_{\ell,b}}{M^{\ell+b}}, \qquad \epsilon = \frac{\max(r,t)}{M}.$$

Note that ϵ is determined by M; it is not a free parameter. By (2), $S_{0,0}/M^0 \le L$ and $S_{0,1}/M^1 \le L$, so $u_0 \le L$, and $v_0 = 0$.

Step 2 (normalised recurrences). Let $1 \le \ell \le d$ and write $u = u_{\ell-1}$, $v = v_{\ell-1}$. By definition, $S'_b \le u\, M^{\ell-1+b}$ and $J'_b \le v\, M^{\ell-1+b}$. Substituting into (3):

$$\begin{aligned}
S_{\ell,1} &\le M^2 \cdot uM^{\ell-1} + r \cdot uM^{\ell} = M^{\ell+1}\Big(1 + \frac{r}{M}\Big)u,\\
S_{\ell,0} &\le t \cdot uM^{\ell-1} + uM^{\ell} + 3vM^{\ell} = M^{\ell}\Big[\Big(1 + \frac{t}{M}\Big)u + 3v\Big],\\
J_{\ell,1} &\le M^2 \cdot vM^{\ell-1} + r \cdot uM^{\ell} = M^{\ell+1}\Big(\frac{r}{M}\,u + v\Big),\\
J_{\ell,0} &\le t \cdot uM^{\ell-1} + 3vM^{\ell} = M^{\ell}\Big(\frac{t}{M}\,u + 3v\Big).
\end{aligned}$$

Dividing by $M^{\ell+1}$, M^{ℓ}, $M^{\ell+1}$, M^{ℓ} respectively, and using $r/M \le \epsilon$, $t/M \le \epsilon$, $v \le 3v$, we get

$$\begin{aligned}
u_\ell &\le (1+\epsilon)u_{\ell-1} + 3v_{\ell-1},\\
v_\ell &\le \epsilon u_{\ell-1} + 3v_{\ell-1}.
\end{aligned} \tag{6}$$

Step 3 (size of ϵ). Since $d \ge 1$, $M \ge 81$, so $\sqrt{M} \ge 9$. Then $r = \lceil \sqrt{M} \rceil \le \sqrt{M} + 1 \le 2\sqrt{M}$ and $t = 16 \le 18 \le 2\sqrt{M}$. Hence, using $\sqrt{M} \ge 3^{d+1}$,

$$\epsilon \le \frac{2\sqrt{M}}{M} = \frac{2}{\sqrt{M}} \le \frac{2}{3^{d+1}}. \tag{7}$$

Step 4 (comparison sequences). Let $U_0 = L$, $V_0 = 0$, and for $\ell \ge 1$

$$U_\ell = (1+\epsilon)U_{\ell-1} + 3V_{\ell-1}, \qquad V_\ell = \epsilon U_{\ell-1} + 3V_{\ell-1}.$$

All coefficients are nonnegative, $u_0 \le U_0$ and $v_0 \le V_0$, so (6) gives $u_\ell \le U_\ell$ and $v_\ell \le V_\ell$ for $0 \le \ell \le d$ by induction on ℓ.

Step 5 (bounding the comparison sequences). We prove by strong induction that, for $0 \le \ell \le d$,

$$U_\ell \le 2L, \qquad V_\ell \le \epsilon L(3^\ell - 1). \tag{8}$$

At $\ell = 0$: $U_0 = L \le 2L$ and $V_0 = 0$. Let $1 \le \ell \le d$ and assume (8) at every index $j < \ell$.

Bound on V_ℓ. Using the bounds at $\ell - 1$,

$$V_\ell = \epsilon U_{\ell-1} + 3V_{\ell-1} \le 2\epsilon L + 3\epsilon L(3^{\ell-1} - 1) = \epsilon L(2 + 3^\ell - 3) = \epsilon L(3^\ell - 1).$$

Bound on U_ℓ. The recursion says $U_{j+1} - U_j = \epsilon U_j + 3V_j$. Summing over $j = 0, \dots, \ell - 1$ and using (8) at the indices $j < \ell$,

$$U_\ell = L + \sum_{j=0}^{\ell-1} (\epsilon U_j + 3V_j) \le L + 2\epsilon L\ell + 3\epsilon L \sum_{j=0}^{\ell-1} (3^j - 1).$$

Since $\sum_{j=0}^{\ell-1}(3^j - 1) = \frac{3^\ell - 1}{2} - \ell$, the right-hand side equals

$$L + \epsilon L\Big[2\ell + \tfrac{3}{2}(3^\ell - 1) - 3\ell\Big] = L + \epsilon L\Big[\tfrac{3}{2}(3^\ell - 1) - \ell\Big].$$

The bracket is positive: $3^\ell = (1+2)^\ell \ge 1 + 2\ell$ (Bernoulli's inequality), so $3^\ell - 1 \ge 2\ell$ and the bracket is at least $3\ell - \ell = 2\ell > 0$. Positivity is what allows us to replace ϵ by its upper bound from (7). Doing so, and using $\ell \le d$,

$$\epsilon\Big[\tfrac{3}{2}(3^\ell - 1) - \ell\Big] \le \frac{2}{3^{d+1}} \cdot \frac{3^{\ell+1} - 3 - 2\ell}{2} = \frac{3^{\ell+1} - 3 - 2\ell}{3^{d+1}} < \frac{3^{\ell+1}}{3^{d+1}} \le 1.$$

Hence $U_\ell < L + L = 2L$. This completes the induction.

Step 6. By Steps 4 and 5, $u_\ell \le U_\ell \le 2L$, i.e. $S_{\ell,b} \le 2LM^{\ell+b}$ for both b. □

5.2 Proof of Theorem 1.1

Proof of Theorem 1.1. Fix an integer $d \geq 1$ and an integer $M \geq 9^{d+1}$, fix a tournament and a labelling as in Section 2 (Lemma 2.1), and form the predicates $P_{\ell,q}$.

Step 1 (the function). Let f be the OR of M disjoint copies of $P_{d,1}$: the input is a tuple $(y^{(1)}, \ldots, y^{(M)})$ of strings in $\{0,1\}^{N_d}$, and $f = 1$ iff $P_{d,1}(y^{(c)}) = 1$ for some c. Each copy has $N_d = LM(kr)^d$ bits, so f is a total Boolean function on

$$n = LM^2(kr)^d \tag{9}$$

bits. By Lemma 3.2 with $\ell = d$, $P_{d,1}(0) = 0$, so $f(0) = 0$.

Step 2 (sensitivity). Let x be an input and e a coordinate; flipping e changes exactly one copy.

- If $f(x) = 0$, every copy rejects, and $f(x^e) = 1$ iff the copy containing e accepts after the flip. Each copy contains at most $S_{d,0}$ such coordinates, so $s(f, x) \leq MS_{d,0}$.
- If $f(x) = 1$ and at least two copies accept, then after any single flip some copy still accepts, so $s(f, x) = 0$.
- If $f(x) = 1$ and exactly one copy accepts, a sensitive coordinate must lie in that copy and turn it from 1 to 0, so $s(f, x) \leq S_{d,1}$.

Lemma 5.1 gives $MS_{d,0} \leq M \cdot 2LM^d$ and $S_{d,1} \leq 2LM^{d+1}$, hence

$$s(f) \leq \max\{MS_{d,0}, S_{d,1}\} \leq 2LM^{d+1}. \tag{10}$$

Step 3 (block sensitivity at zero). By Lemma 3.2 with $\ell = d$, there are Mk^d pairwise disjoint nonempty blocks in $[N_d]$, each of which turns $P_{d,1}$ from 0 to 1 when flipped at the all-zero input. Place this family in each of the M copies. The resulting M^2k^d blocks are pairwise disjoint, and flipping any one of them at 0 makes one copy accept, so f becomes $1 \neq f(0)$. Thus

$$\mathrm{bs}(f, 0) \geq M^2k^d. \tag{11}$$

In particular f is nonconstant. Hence $s(f) > 0$: the hypercube is connected, so a nonconstant function differs across some edge.

Step 4 (ratio). Combining (10) and (11), and using $k/M^2 = 2 + 1/M^2 \geq 2$ and $L = d + 2$,

$$\frac{\mathrm{bs}(f)}{s(f)^2} \geq \frac{\mathrm{bs}(f, 0)}{s(f)^2} \geq \frac{M^2k^d}{4L^2M^{2d+2}} = \frac{(k/M^2)^d}{4L^2} \geq \frac{2^d}{4(d+2)^2}. \tag{12}$$

This is the second assertion of the theorem.

Step 5 (arbitrary C). The right-hand side of (12) tends to infinity with d. Given $C > 0$, choose d with $2^d/(4(d+2)^2) > C$, then $M \geq 9^{d+1}$, then the labelling, and form f. Then $\mathrm{bs}(f) > C\, s(f)^2$. □

6 A fixed exponent by composition

The construction supplies a fixed seed whose block sensitivity at zero exceeds the square of its sensitivity; iterated composition of the seed gives Corollary 1.2. The next lemma contains the two composition facts needed.

Lemma 6.1. *Let $f : \{0,1\}^{n_1} \to \{0,1\}$ and $g : \{0,1\}^{n_2} \to \{0,1\}$ be total Boolean functions. Their disjoint composition is the function on n_1n_2 bits defined by*

$$(f \circ g)(x_1, \ldots, x_{n_1}) = f(g(x_1), \ldots, g(x_{n_1})), \qquad x_i \in \{0,1\}^{n_2}.$$

Then

$$s(f \circ g) \le s(f)s(g).$$

If $g(0) = 0$*, then* $(f \circ g)(0) = f(0)$ *and*

$$\mathrm{bs}(f \circ g, 0) \ge \mathrm{bs}(f, 0)\,\mathrm{bs}(g, 0).$$

More explicitly, any disjoint families of p *and* p' *blocks sensitive at zero for* f *and* g*, respectively, yield a disjoint family of* pp' *blocks sensitive at zero for* $f \circ g$*.*

Proof. Identify the coordinates of the composition with $[n_1] \times [n_2]$: coordinate (i, e) is coordinate e of the block x_i.

Sensitivity. Fix $x = (x_1, \ldots, x_{n_1})$ and put $w = (g(x_1), \ldots, g(x_{n_1})) \in \{0,1\}^{n_1}$, so $(f \circ g)(x) = f(w)$. Flipping coordinate (i, e) replaces x_i by x_i^e and changes nothing else. If $g(x_i^e) = g(x_i)$, the outer string is still w and the value is unchanged. If $g(x_i^e) \ne g(x_i)$, the outer string becomes w^i, and the value changes iff $f(w^i) \ne f(w)$. So (i, e) is sensitive for $f \circ g$ at x exactly when e is sensitive for g at x_i and i is sensitive for f at w. Therefore

$$s(f \circ g, x) = \sum_{i:\, f(w^i) \ne f(w)} s(g, x_i) \le s(f, w)\, s(g) \le s(f)s(g).$$

Blocks at zero. Suppose $g(0) = 0$. Then $(f \circ g)(0) = f(g(0), \ldots, g(0)) = f(0)$. Let $X_1, \ldots, X_p \subseteq [n_1]$ be pairwise disjoint sensitive blocks for f at 0, and $Y_1, \ldots, Y_{p'} \subseteq [n_2]$ pairwise disjoint sensitive blocks for g at 0. For $1 \le s \le p$ and $1 \le s' \le p'$ set

$$Z_{s,s'} = X_s \times Y_{s'} \subseteq [n_1] \times [n_2].$$

- *Nonempty and pairwise disjoint.* Each factor is nonempty. If $(s, s') \ne (\sigma, \sigma')$, then either $s \ne \sigma$ and $X_s \cap X_\sigma = \emptyset$, or $s = \sigma$, $s' \ne \sigma'$ and $Y_{s'} \cap Y_{\sigma'} = \emptyset$; in both cases the products are disjoint.
- *Sensitive.* In $0^{Z_{s,s'}}$, the block x_i equals $0^{Y_{s'}}$ for $i \in X_s$ and equals 0 for $i \notin X_s$. Since $g(0) = 0$ and $Y_{s'}$ is sensitive for g at 0, $g(0^{Y_{s'}}) = 1$. So the outer string is 1 exactly on X_s, i.e. it is the indicator string 0^{X_s} of X_s. Since X_s is sensitive for f at 0, $f(0^{X_s}) \ne f(0) = (f \circ g)(0)$.

So the pp' sets $Z_{s,s'}$ are pairwise disjoint sensitive blocks for $f \circ g$ at 0. Taking $p = \mathrm{bs}(f, 0)$ and $p' = \mathrm{bs}(g, 0)$ gives the inequality; if either is zero it is trivial. □

Proof of Corollary 1.2. Step 1 (the seed). Fix $d = 9$, choose an integer $M \ge 9^{10}$ and an admissible labelling, and let f be the function constructed in the proof of Theorem 1.1 for these parameters. Here $L = 11$ and $k = 2M^2 + 1$. Put

$$A = 22M^{10}, \qquad B = M^2 k^9.$$

By (10), $s(f) \le 2 \cdot 11 \cdot M^{10} = A$; by (11), $\mathrm{bs}(f, 0) \ge B$; and $f(0) = 0$ (Step 1 of that proof).

Step 2 (the exponent). Since $k/M^2 \ge 2$,

$$\frac{B}{A^2} = \frac{M^2 k^9}{484\, M^{20}} = \frac{(k/M^2)^9}{484} \ge \frac{2^9}{484} = \frac{128}{121} > 1.$$

Also $A > 1$. Hence $B > A^2 > 1$ and

$$\alpha = \frac{\log B}{\log A} > 2.$$

This α depends on the chosen M and is fixed once M is. By definition, $A^\alpha = B$.

Step 3 (iteration). Set $F_1 = f$ and $F_{\nu+1} = f \circ F_\nu$ for $\nu \geq 1$. With n as in (9), F_ν is a total function on n^ν bits. We show by induction on ν that

$$F_\nu(0) = 0, \qquad s(F_\nu) \leq A^\nu, \qquad \mathrm{bs}(F_\nu, 0) \geq B^\nu.$$

For $\nu = 1$ this is Step 1. If it holds for ν, apply Lemma 6.1 with outer function f and inner function $g = F_\nu$, which satisfies $g(0) = 0$: $F_{\nu+1}(0) = f(0) = 0$, $s(F_{\nu+1}) \leq s(f)s(F_\nu) \leq A \cdot A^\nu$, and $\mathrm{bs}(F_{\nu+1}, 0) \geq \mathrm{bs}(f, 0)\,\mathrm{bs}(F_\nu, 0) \geq B \cdot B^\nu$.

Step 4 (conclusions). Since $\mathrm{bs}(F_\nu, 0) \geq B^\nu \geq 1$, there is a block whose flip at 0 changes F_ν, so F_ν is nonconstant. Since $B > 1$, $\mathrm{bs}(F_\nu, 0) \to \infty$. Since $t \mapsto t^\alpha$ is increasing on $[0, \infty)$ and $A^\alpha = B$,

$$s(F_\nu)^\alpha \leq (A^\nu)^\alpha = B^\nu \leq \mathrm{bs}(F_\nu, 0).$$

Finally, $s(F_\nu) > 0$ because F_ν is nonconstant, and the same estimates give the explicit ratio bound

$$\frac{\mathrm{bs}(F_\nu)}{s(F_\nu)^2} \geq \frac{\mathrm{bs}(F_\nu, 0)}{s(F_\nu)^2} \geq \left(\frac{B}{A^2}\right)^\nu \geq \left(\frac{128}{121}\right)^\nu. \qquad \square$$

7 Verification checklist

This document contains no "Unverified step" boxes: each step of Sections 2–6 was rechecked by hand (not by machine) during the rewrite, and none raised a doubt. That is a statement about the rewriting process, not evidence; the reader should check independently. The background of Section 1.2 was not checked and is not used. A sceptical reader should examine the following first.

1. **One flip repairs exactly one failed condition** (Lemma 2.3(b)–(d)). Everything in Section 4 rests on a clause testing each child at most once. Check that gate children of one clause lie in distinct rows, all different from the target row.

2. **Directions of nesting.** (N1)/(N0) are applied to the child indices $Q - q < Q$ and $\gamma' = Q - q' < \gamma = Q - q$. Check each use: Step B ((4) and the gate case after it), C2 (a passing gate forces a failed target), C3(i) (a satisfied target forces a failed gate), D2, E2.

3. $|\mathcal{T}| \leq 15$ (Step C2 with Lemma 2.1). Check that the derived relation $a(i, j) = m_j$ is exactly the configuration excluded by the lemma, and the arithmetic $\binom{16}{2} = 120$, $k^{16} r^{16-120} \leq 3^{16} M^{-20} < 1$.

4. $|\mathcal{G}| \leq 3$ **and the charges** (Steps C3, D2, D3). Check that only non-exceptional clauses use (5), that at most one clause is exceptional, and that the total is $S_1' + 3J_1'$.

5. **No S_1' term in $J_{\ell,0}$** (Step E2). Check that the initial values $P_{\ell-1,\gamma} = P_{\ell-1,\gamma'} = 1$ use only the failed gate at index q and nesting, and that the witness is chosen at the larger index q' (Step E0).

6. **Powers of M in the normalisation** (Step 2 of Lemma 5.1), and the base values $u_0 \leq L$, $v_0 = 0$ from Lemma 3.1 (in particular Step 4 there, $J_{0,b} = 0$).

7. **The induction $U_\ell < 2L$** (Step 5 of Lemma 5.1). The hypothesis $M \geq 9^{d+1}$ is used only to obtain (7) (Step 3), and (7) is used only here, together with $\ell \leq d$.

8. **The final count** (proof of Theorem 1.1). Check $s(f) \leq \max\{MS_{d,0}, S_{d,1}\}$, $\mathrm{bs}(f, 0) \geq M^2 k^d$ from Lemma 3.2, and $k/M^2 \geq 2$ in (12).

9. **Composition** (Lemma 6.1). Check that the block bound is only claimed at the all-zero input and needs $g(0) = 0$, which the induction in the Corollary maintains, and that $2^9/484 = 128/121 > 1$.

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